

CONNECTING & SHAPING DATA

FRONT-END VS. BACK-END

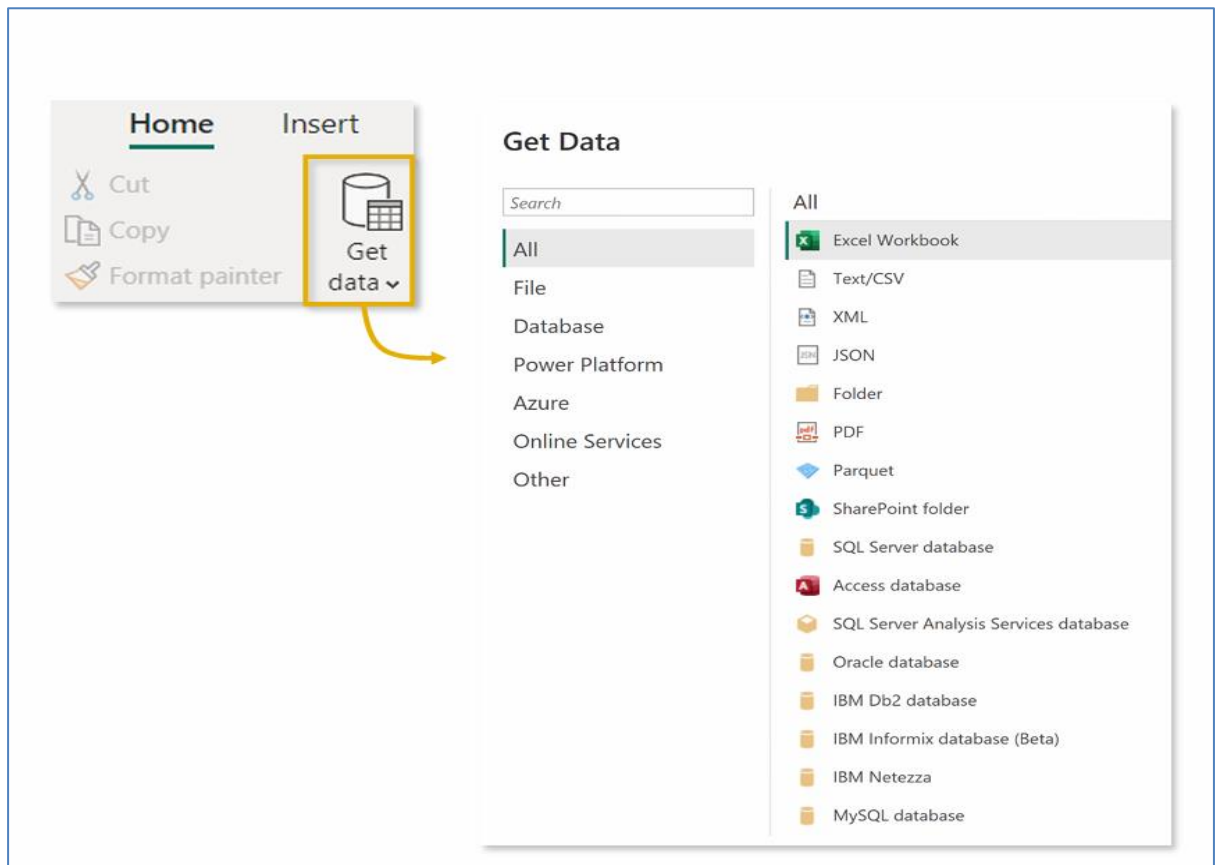
Power BI Desktop essentially has two distinct environments: a **front-end** and a **back-end**

- The **front-end** includes the **Data, Model & Report** views, where most of the modeling, analysis and visualization takes place
- The **back-end** includes the **Power Query Editor**, where raw data is extracted, transformed, and loaded to the front-end (ETL)

TYPES OF DATA CONNECTORS

Power BI can connect to virtually any type of source data, including (but not limited to):

- Flat files & Folders (csv, text, xlsx, etc.)
- Databases (SQL, Access, Oracle, IBM, etc.)
- Power Platform (Datasets, DataMart's, Dataflows, Dataverse, etc.)
- Azure (Azure SQL, Analysis Services, Databricks, etc.)
- Online Services (SharePoint, GitHub, Dynamics 365, Google Analytics, Salesforce, Power BI Service, etc.)
- Other (Web feeds, R scripts, Spark, Hadoop, etc.)



POWER QUERY EDITOR

The screenshot shows the Power Query Editor interface with several key components highlighted and labeled:

- Query Editing Tools** (Table transformations, calculated columns, etc.): Points to the ribbon tabs (File, Home, Insert, Modeling, View, Optimize, Help, External Tools) and the ribbon groups (General, Transform, Add Column, View, Tools, Help).
- Formula Bar** (this is "M" code): Points to the formula bar at the top of the main workspace.
- Table Name & Properties**: Points to the "Properties" pane on the right, which shows the table name and its properties.
- Applied Steps** (like a macro): Points to the "Applied Steps" pane on the right, which lists the steps applied to the query.
- Queries Pane** (list of all queries): Points to the "Queries" pane on the left, which lists all queries in the data source.
- Table Preview**: Points to the main workspace area showing a preview of the data table.

QUERY EDITING TOOLS

The **HOME** tab includes **general settings** and **common table transformation tools**

The **TRANSFORM** tab includes tools to **modify existing columns** (splitting/grouping, transposing, extracting text, etc.)

The **ADD COLUMN** tools **create new columns** (based on conditional rules, text operations, calculations, dates, etc.)

BASIC TABLE TRANSFORMATIONS

Sort values (A-Z, Low-High, etc.)

Change data type (date, \$, %, text, etc.)

Promote headers

Choose or remove columns

Remove Columns

Remove Other Columns

Tip: use the "Remove Other Columns" option if you always want a specific set

Remove Top Rows

Remove Bottom Rows

Remove Alternate Rows

Remove Duplicates

Remove Blank Rows

Remove Errors

Keep or remove rows

Tip: use "Remove Duplicates" to create new lookup tables from scratch

Duplicate, move or rename columns

Tip: Right-click column headers to access common tools

Copy

Remove

Remove Other Columns

Duplicate Column

Add Column From Examples...

Remove Duplicates

Remove Errors

Change Type

Transform

Replace Values...

Replace Errors...

Fill

Unpivot Columns

Unpivot Other Columns

Unpivot Only Selected Columns

Rename...

Move

Drill Down

Add as New Query

CONNECTING TO A DATABASE

Get Data & enter credentials

MySQL database

Server

127.0.0.1

Database

mavenfuzzyfactory

SQL statement (optional, requires database)

Include relationship columns

Navigate using full hierarchy

OK

Cancel

Write custom or advanced queries with SQL statements (optional)

Select tables & transform

Power Query - Choose data

mavenfuzzyfactory

mavenfuzzyfactory.website_sessions

12 website_session_id

12 created_at

12 user_id

12 is_request_session

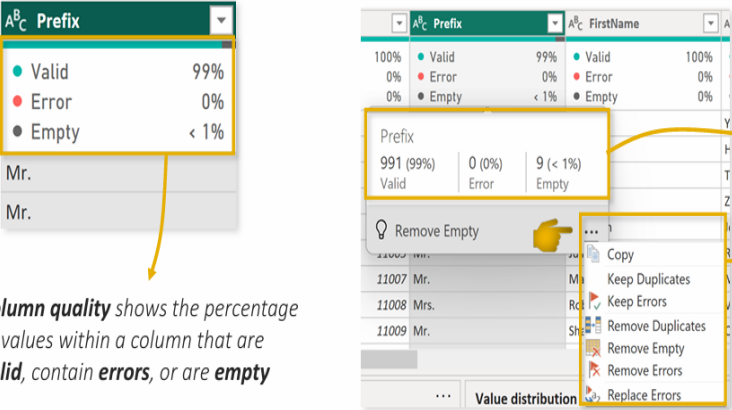
12 utm_source

12 utm_campaign

Cancel

Transform data

DATA PROFILING: COLUMN QUALITY



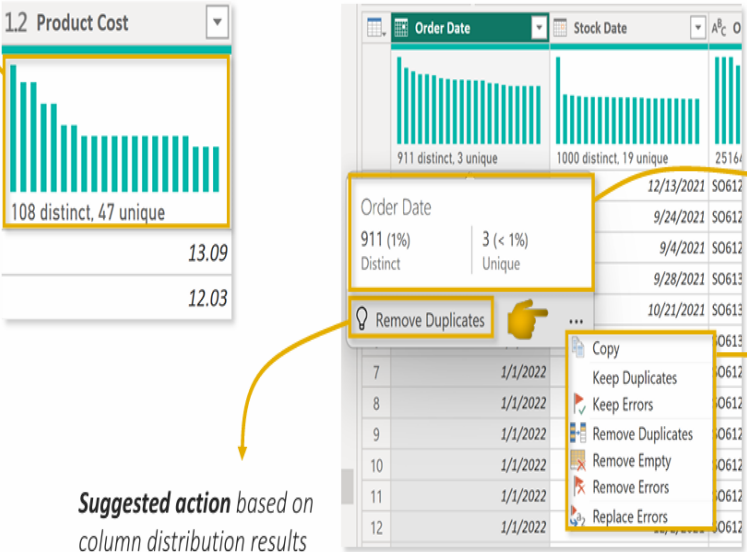
Column quality shows the percentage of values within a column that are **valid**, contain **errors**, or are **empty**

Hover over the column quality box to see the **number of records** in each category

Click the **options menu** to remove duplicates, errors or empty values

DATA PROFILING: COLUMN DISTRIBUTION

Column distribution provides a sample distribution of the data in a column



Hover over the column quality box to see the **number of distinct & unique records**

Click the **options menu** to remove duplicates, errors or empty values

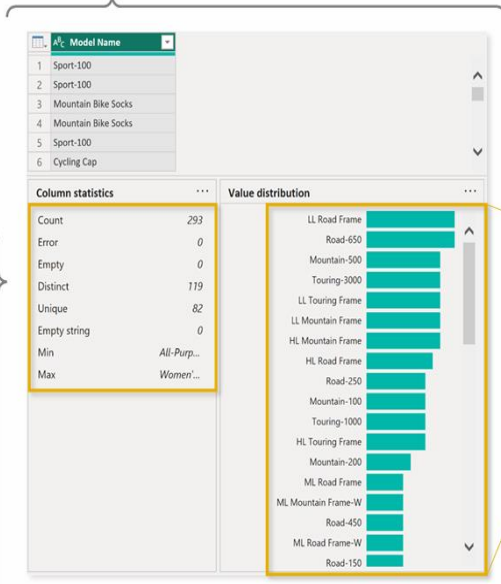
Suggested action based on column distribution results

DATA PROFILING: COLUMN PROFILE

Column profile provides a more holistic view of the data in a column, including a sample distribution and profiling statistics

Column statistics provide more detailed profiling metrics, including:

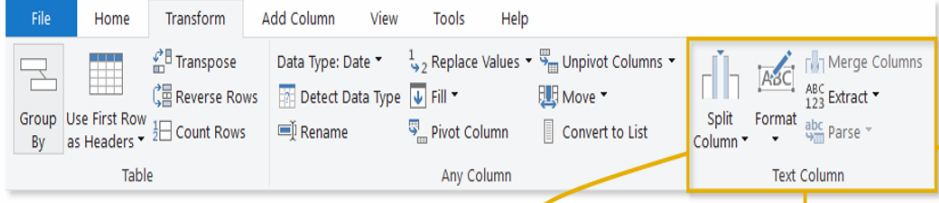
- Count = 293**
(total number of values in column)
- Distinct Count = 119**
(total number of distinct values, whether they appear once or multiple times)
- Unique = 82**
(total number of values that appear exactly once)
- Min & Max**
(lowest and highest observed values)
Note: Typically only useful for numerical values



The screenshot shows the 'Model Name' column profile. The 'Column statistics' panel lists: Count (293), Error (0), Empty (0), Distinct (119), Unique (82), Empty string (0), Min (All-Purp...), and Max (Women...). The 'Value distribution' panel shows a horizontal bar chart for various model names like 'LL Road Frame', 'Road-650', 'Mountain-500', etc. A callout box highlights the 'Date_and_Time' value '2015-01-14T06:36:34' with a dropdown menu showing 'Equals', 'Does Not Equal', and 'Copy' options.

Hover over the value distribution bar for **suggested transformations** and additional options

TEXT TOOLS



The screenshot shows the 'Transform' ribbon in Power Query. The 'Text' group contains 'Split Column', 'Format', and 'Merge Columns'. A callout box highlights the 'Split Column' dropdown menu with options: 'By Delimiter', 'By Number of Characters', 'By Positions', 'By Lowercase to Uppercase', 'By Uppercase to Lowercase', 'By Digit to Non-Digit', and 'By Non-Digit to Digit'.

Split a text column based on a specific delimiter, number of characters, or other attributes

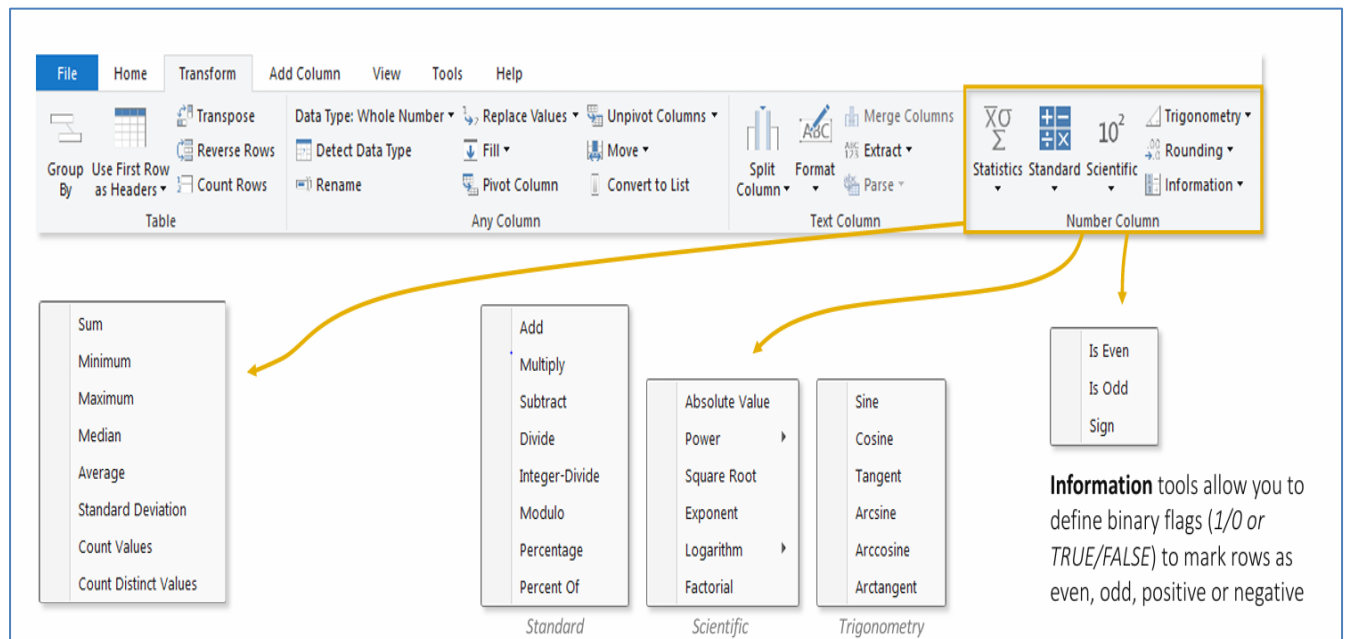
Extract characters from text based on fixed lengths, first/last characters, ranges or delimiters

The 'Extract' dropdown menu shows options: 'Length', 'First Characters', 'Last Characters', 'Range', 'Text Before Delimiter', 'Text After Delimiter', and 'Text Between Delimiters'.

The 'Format' dropdown menu shows options: 'lowercase', 'UPPERCASE', 'Capitalize Each Word', 'Trim', 'Clean', 'Add Prefix', and 'Add Suffix'.

You can access many tools from both the **Transform** and **Add Column** menus - the difference is whether you want to **ADD** a new column or **OVERWRITE** an existing one.

NUMERICAL TOOLS:



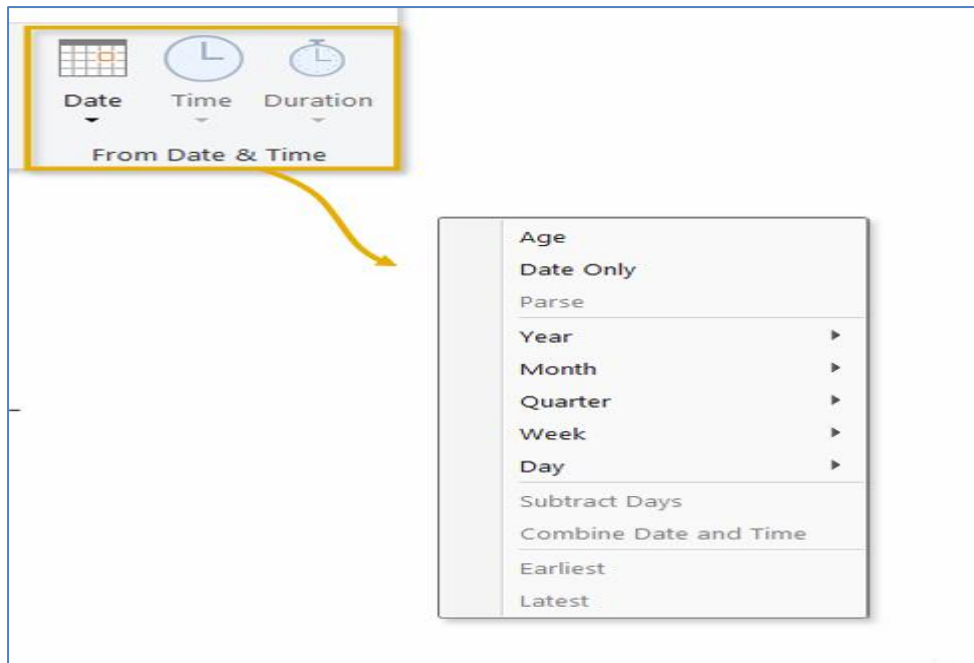
Statistics functions allow you to evaluate basic stats for a selected column (sum, min/max, average, count, count distinct, etc.)

Standard, Scientific and **Trigonometry** tools allow you to apply standard operations (addition, multiplication, division, etc.) or more advanced calculations (power, logarithm, sine, tangent, etc.) to each value in a column

DATE & TIME TOOLS

Date & Time tools are relatively straight-forward, and include the following options:

- **Age:** Difference between the current date and the date in each row
- **Date Only:** Removes the time component from a date/time field
- **Year/Month/Quarter/Week/Day:** Extracts individual components from a date field (time specific options include Hour, Minute, Second, etc.)
- **Earliest/Latest:** Evaluates the earliest or latest date from a column as a single value (can only be accessed from the "Transform" menu)



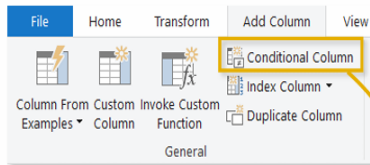
INDEX COLUMNS

Index Columns contain a list of sequential values that can be used to identify each unique row in a table (*typically starting from 0 or 1*)

These are often used to create **unique IDs** that can be used to form relationships between tables (*more on that later!*)

	I_3 Index	Order Date	Stock Date	A_6^c Order Number	I_3 Product Key
1	1	1/1/2020	9/21/2019	SO45080	332
2	2	1/1/2020	12/5/2019	SO45079	312
3	3	1/1/2020	10/29/2019	SO45082	350
4	4	1/1/2020	11/16/2019	SO45081	338
5	5	1/2/2020	12/15/2019	SO45083	312
6	6	1/2/2020	10/12/2019	SO45084	310
7	7	1/2/2020	12/18/2019	SO45086	314
8	8	1/2/2020	10/9/2019	SO45085	312
9	9	1/3/2020	10/3/2019	SO45093	312
10	10	1/3/2020	9/29/2019	SO45090	310
11	11	1/3/2020	12/11/2019	SO45088	345
12	12	1/3/2020	10/24/2019	SO45092	313
13	13	1/3/2020	12/16/2019	SO45089	351
14	14	1/3/2020	10/26/2019	SO45091	314
15	15	1/3/2020	9/11/2019	SO45087	350
16	16	1/3/2020	9/11/2019	SO45094	310
17	17	1/4/2020	10/30/2019	SO45096	312
18	18	1/4/2020	10/30/2019	SO45097	313
19	19	1/4/2020	9/15/2019	SO45098	310
20	20	1/4/2020	12/7/2019	SO45095	344

CONDITIONAL COLUMNS



Conditional Columns allow you to define new fields based on logical rules and conditions (IF/THEN statements)

Here we're creating a conditional column named **Quantity Type**, which is based on **Order Quantity**:

- If Order Quantity =1, Quantity Type = "Single Item"
- Else If Order Quantity >1, Quantity Type = "Multiple Items"
- Else; Quantity Type = "Other"

Add Conditional Column

Add a conditional column that is computed from the other columns or values.

New column name
QuantityType

	Column Name	Operator	Value		Output
If	Order Quantity	equals	1	Then	Single Item
Else If	Order Quantity	is greater than	1	Then	Multiple Items
Add Clause					
Else					Other

[OK](#) [Cancel](#)



Data Source



Power Query



Power BI Front-End

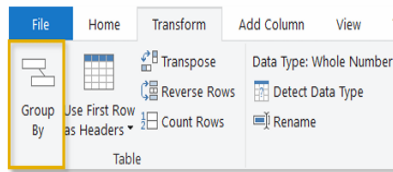


Published Reports

UPSTREAM

DOWNSTREAM

GROUPING & AGGREGATING



Group By allows you to aggregate data at a different level or “grain” (i.e. group daily records into monthly, aggregate transactions by store, etc.)

	Order Date	Product Key	Customer Key	Order Quantity
1	6/25/2022	214	14719	1
2	10/6/2021	214	21990	1
3	12/30/2021	214	22098	1
4	6/29/2022	214	22748	1
5	8/16/2021	214	27821	1
6	10/9/2021	214	15685	1
7	8/9/2021	214	14951	1
8	1/19/2022	214	23101	1
9	9/23/2021	214	17158	1
10	1/19/2022	214	24196	1
11	6/29/2022	214	12963	1
12	9/13/2021	214	12715	1
13	10/2/2021	214	14846	1
14	7/31/2021	214	11290	1
15	11/24/2021	214	22103	1
16	8/1/2021	214	16982	1
17	10/12/2021	214	20410	1
18	9/10/2021	214	14217	1
19	10/12/2021	214	19642	1
20	8/11/2021	214	11666	1

Group By

Specify the column to group by and the desired output.

☒ Basic
 ☐ Advanced

Product Key

New column name
TotalQuantity

Operation
Sum

Column
Order Quantity

OK
 Cancel

	Product Key	TotalQuantity
1	214	2099
2	215	1940
3	220	1985
4	223	4151
5	226	392
6	229	408
7	232	424
8	235	381
9	310	169
10	311	139
11	312	179
12	313	168
13	314	157
14	320	65
15	322	39
16	324	72
17	326	65

Here we're transforming a daily, transaction-level table into a summary of **Total Quantity** by **Product Key**

NOTE: Any fields not specified in the Group By settings are lost

	Order Date	Product Key	Customer Key	Order Quantity
1	6/25/2022	214	14719	1
2	10/6/2021	214	21990	1
3	12/30/2021	214	22098	1
4	6/29/2022	214	22748	1
5	8/16/2021	214	27821	1
6	10/9/2021	214	15685	1
7	8/9/2021	214	14951	1
8	1/19/2022	214	23101	1
9	9/23/2021	214	17158	1
10	1/19/2022	214	24196	1
11	6/29/2022	214	12963	1
12	9/13/2021	214	12715	1
13	10/2/2021	214	14846	1
14	7/31/2021	214	11290	1
15	11/24/2021	214	22103	1
16	8/1/2021	214	16982	1
17	10/12/2021	214	20410	1
18	9/10/2021	214	14217	1
19	10/12/2021	214	19642	1
20	8/11/2021	214	11666	1

Group By

Specify the columns to group by and one or more outputs.

☐ Basic
 ☒ Advanced

Product Key

Customer Key

Add grouping

New column name
TotalQuantity

Operation
Sum

Column
Order Quantity

Add aggregation

OK
 Cancel

	Product Key	Customer Key	TotalQuantity
1	214	29358	1
2	214	15101	1
3	214	12473	1
4	214	12963	1
5	214	26989	1
6	214	13203	1
7	214	14951	1
8	214	11201	1
9	214	19538	1
10	214	22749	1
11	214	15815	1
12	214	19252	1
13	214	14849	1
14	214	11290	1
15	214	27851	1
16	214	16982	1
17	214	21963	1
18	214	19725	1
19	214	15684	1
20	214	11666	1
21	214	26941	1

This time we're transforming the daily, transaction-level table into a summary of **Total Quantity** grouped by both **Product Key** and **Customer Key** (using the “Advanced” option)

NOTE: This is like creating a PivotTable in Excel and pulling in **Sum of Order Quantity** with **Product Key** and **Customer Key** as row labels

PIVOTING & UNPIVOTING

Pivoting describes the process of turning distinct row values into columns, and unpivoting describes the process of turning distinct columns into rows

Imagine the table on a hinge; **pivoting** rotates it from **vertical** to **horizontal**, and **unpivoting** rotates it from **horizontal** to **vertical**

NOTE: *Transpose* works very similarly, but doesn't recognize unique values; instead, the entire table is transformed so that each row becomes a column and vice versa

MERGING QUERIES

Merge Queries

Merge

Select a table and matching columns to create a merged table.

Sales Data

Order Date	Product Key	Customer Key	Order Quantity	Index	Stock Date	Order Number	Territory
6/25/2022	214	14719	1	55115	4/20/2022	SO73780	
10/8/2021	214	21990	1	14247	7/2/2021	SO55746	
12/30/2021	214	22098	1	26322	11/10/2021	SO61052	
6/29/2022	214	22748	1	55740	4/9/2022	SO74069	

Product Lookup

Product Key	Product Subcategory Key	Product S K U	Product Name	Model Name	
214	31	HL-U509-R	Sport-100 Helmet, Red	Sport-100	Universal fit, v
215	31	HL-U509	Sport-100 Helmet, Black	Sport-100	Universal fit, v
218	23	SO-B909-M	Mountain Bike Socks, M	Mountain Bike Socks	Combination c
219	23	SO-B909-L	Mountain Bike Socks, L	Mountain Bike Socks	Combination c

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

> Fuzzy matching options

✓ The selection matches 56046 of 56046 rows from the first table.

OK Cancel

APPENDING QUERIES

Append

Concatenate rows from two tables into a single table.

☒ Two tables ☐ Three or more tables

First table
AdventureWorks Sales Data 2020

Second table
AdventureWorks Sales Data 2021

APPENDING FILES FROM A FOLDER

Get Data

Search

All

Excel Workbook

Text/CSV

XML

JSON

Folder

PDF

Parquet

SharePoint folder

SQL Server database

Access database

SQL Server Analysis Services database

Oracle database

IBM Db2 database

IBM Informix database (Beta)

IBM Netezza

MySQL database

Connect Cancel

Folder

Folder path
C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales

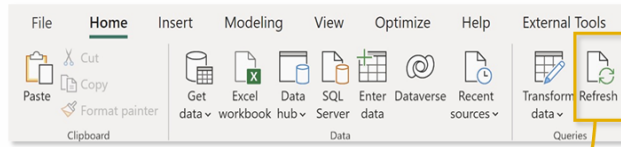
OK Cancel

C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales

Content	Name	Extension	Date accessed	Date modified	Date created	Attributes	C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales
Binary	AdventureWorks Sales Data 2020.csv	.csv	12/11/2022 6:17:52 PM	12/10/2022 4:09:09 PM	12/11/2022 6:17:52 PM	Record	C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales
Binary	AdventureWorks Sales Data 2021.csv	.csv	12/11/2022 6:17:52 PM	12/10/2022 4:09:09 PM	12/11/2022 6:17:52 PM	Record	C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales
Binary	AdventureWorks Sales Data 2022.csv	.csv	12/11/2022 6:17:52 PM	12/10/2022 7:00:24 PM	12/11/2022 6:17:52 PM	Record	C:\Users\Branislav Poljasevic\Documents\3. PowerBI Desktop\Sales

Combine Load Transform Data Cancel

REFRESHING QUERIES



By default, **all queries** will refresh when you use the **Refresh** command from the **Home** tab

From the Query Editor, uncheck **Include in report refresh** to exclude individual queries from the refresh



PRO TIP: Exclude queries from refresh that don't change often (like lookups or static data tables)

